

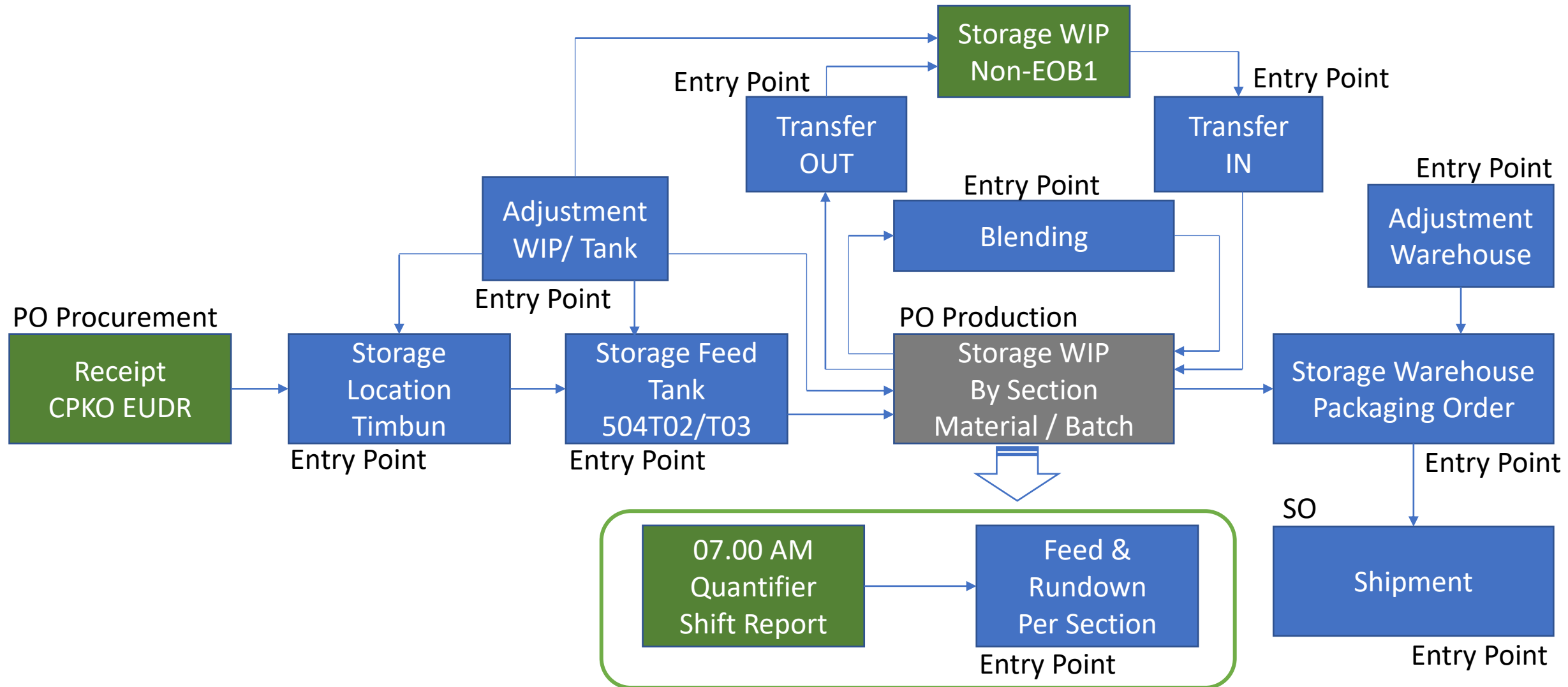
EUDR – Traceability System (EUDR – TS) Plant EOB1

User Guidance

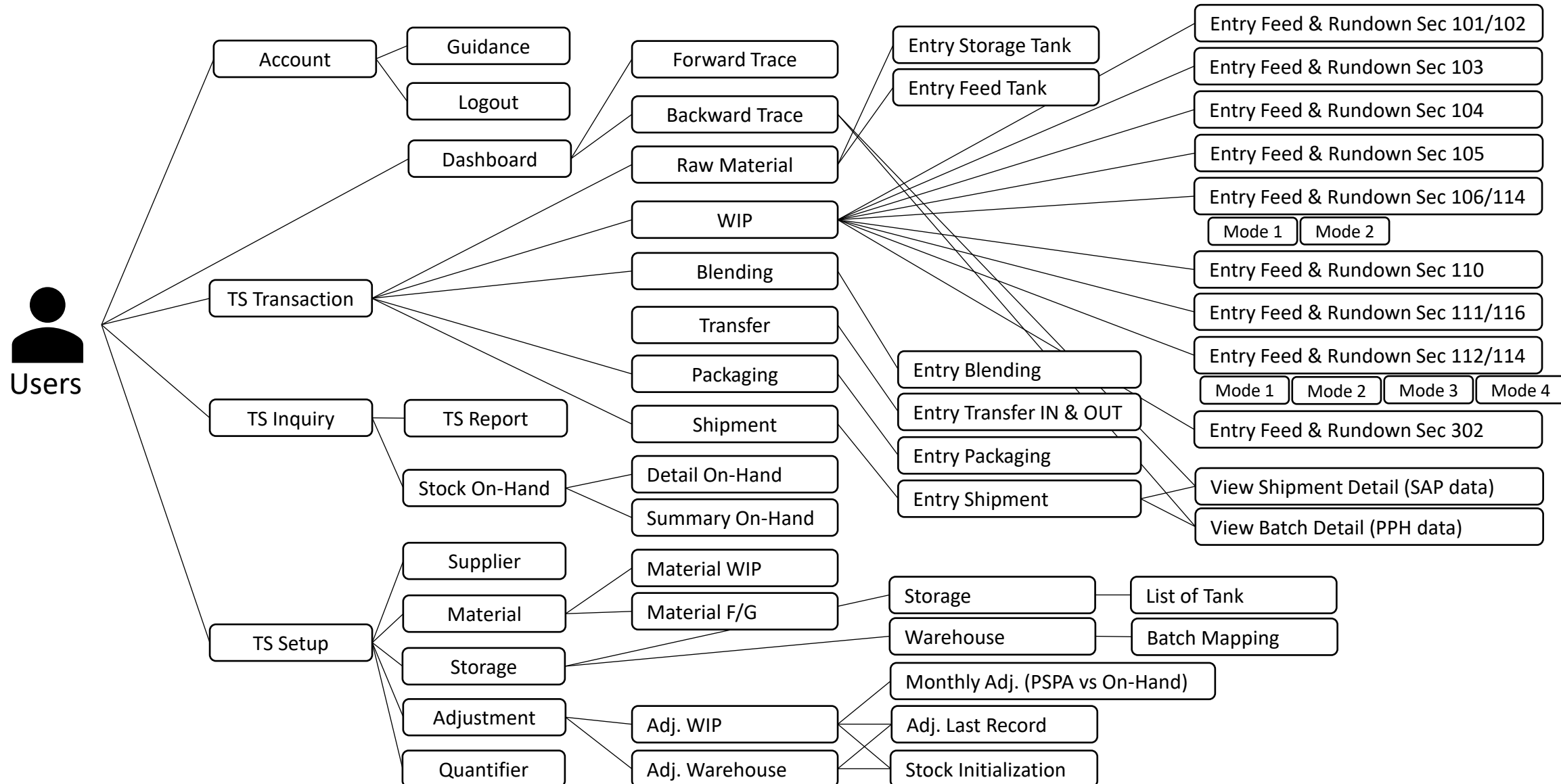
(ver. 20250328)

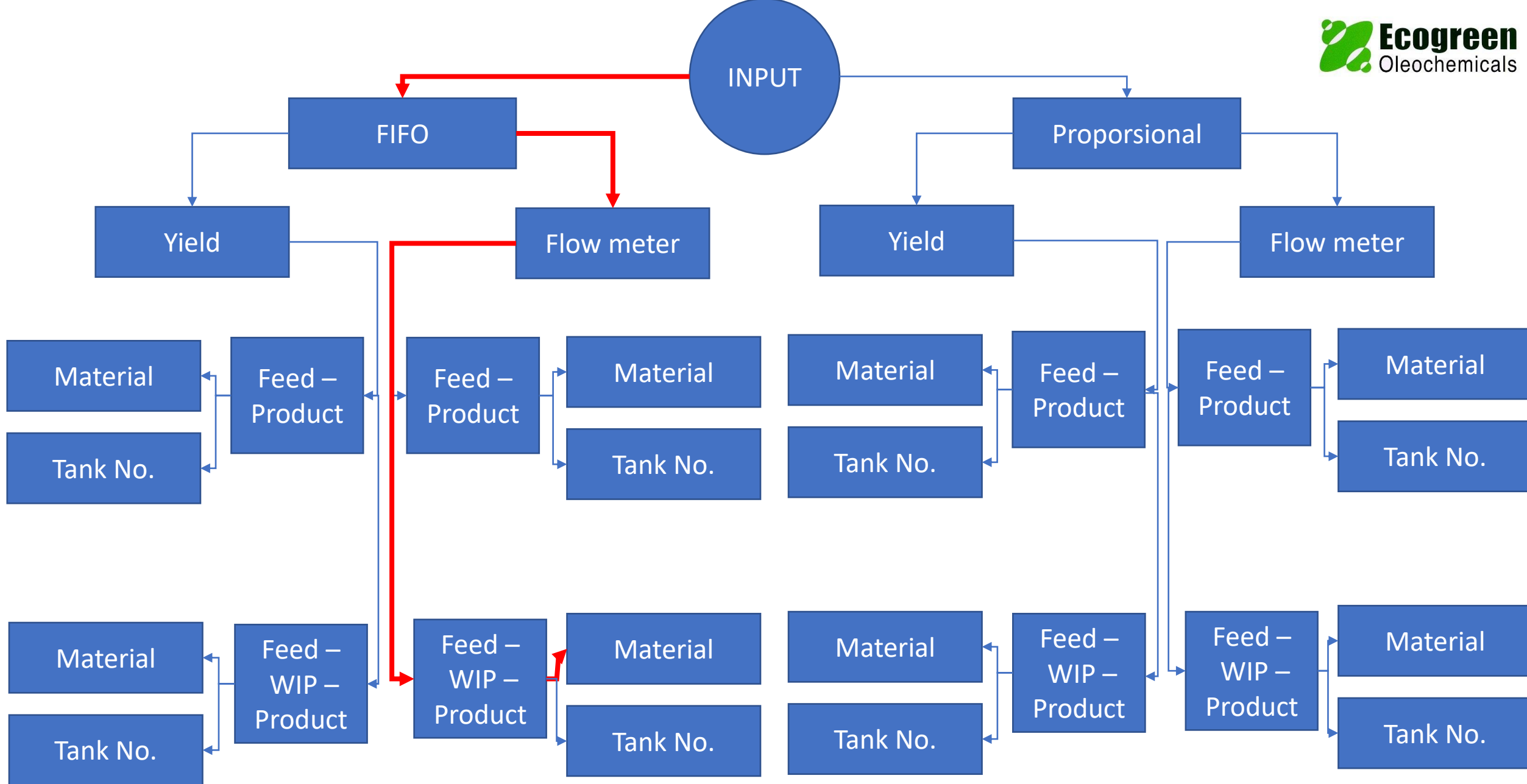
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Business Process EOB1 EUDR-TS

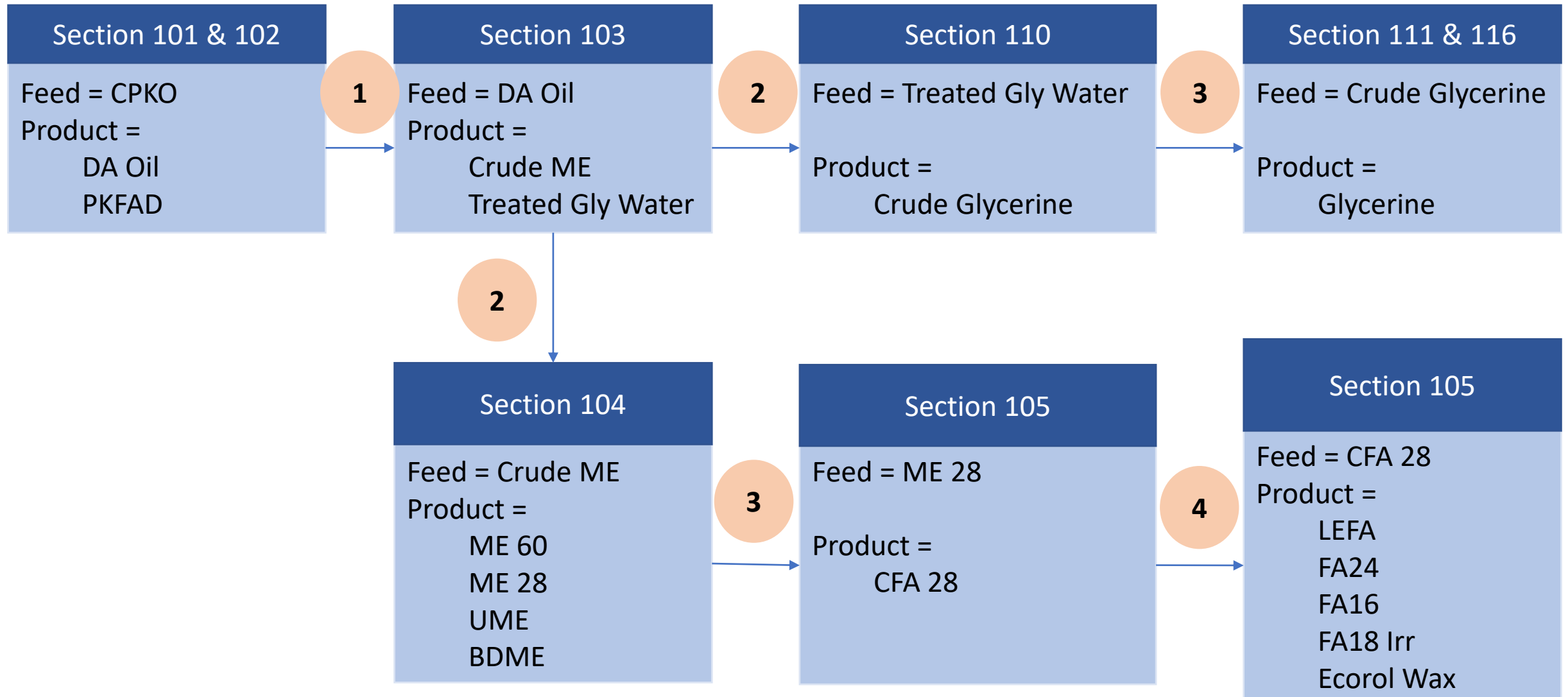


EUDR-TS Modules

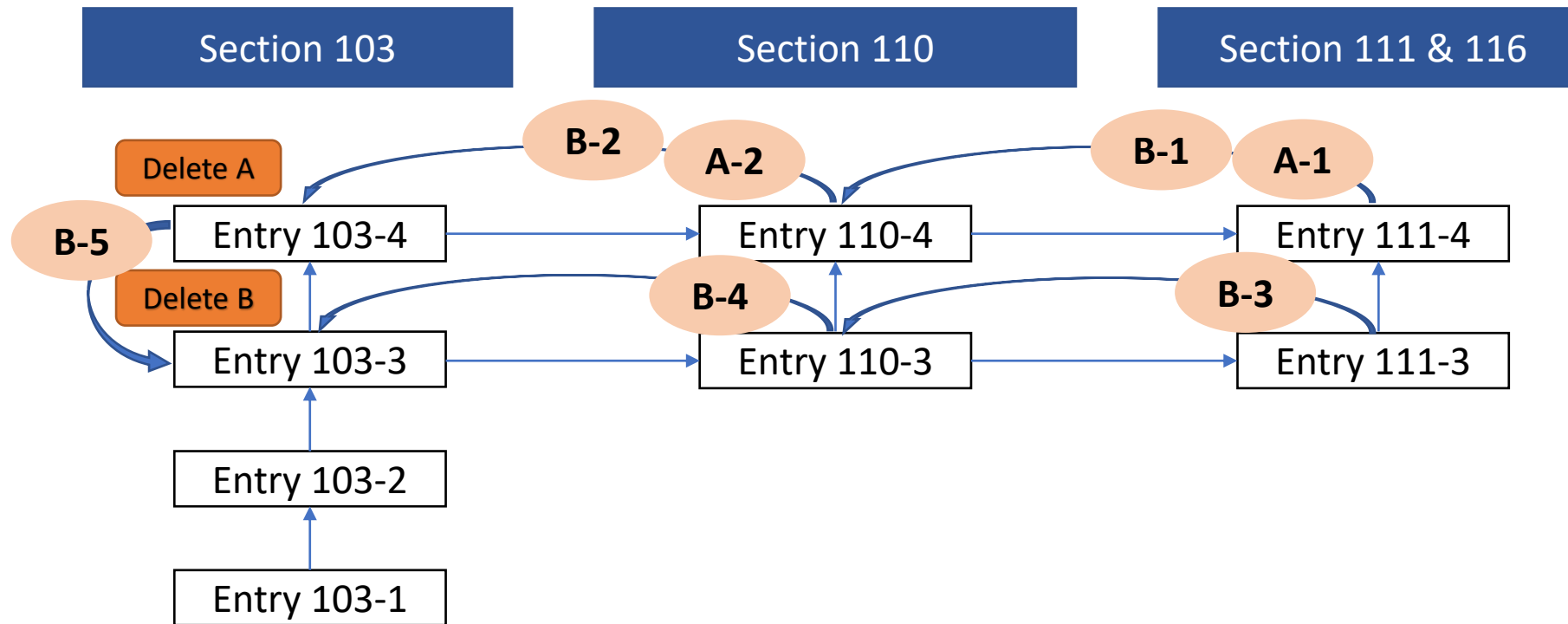




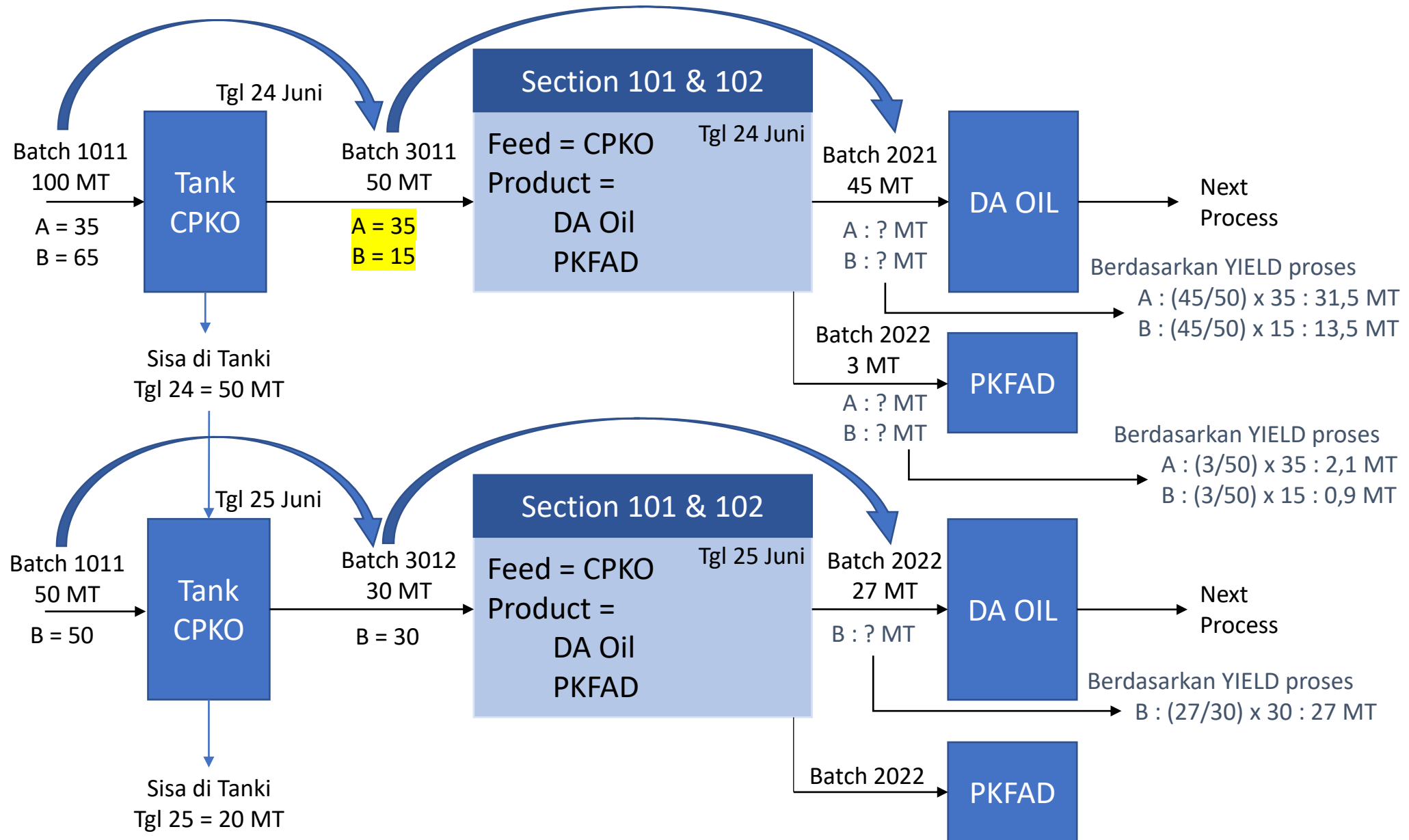
Input Sequential in the Process



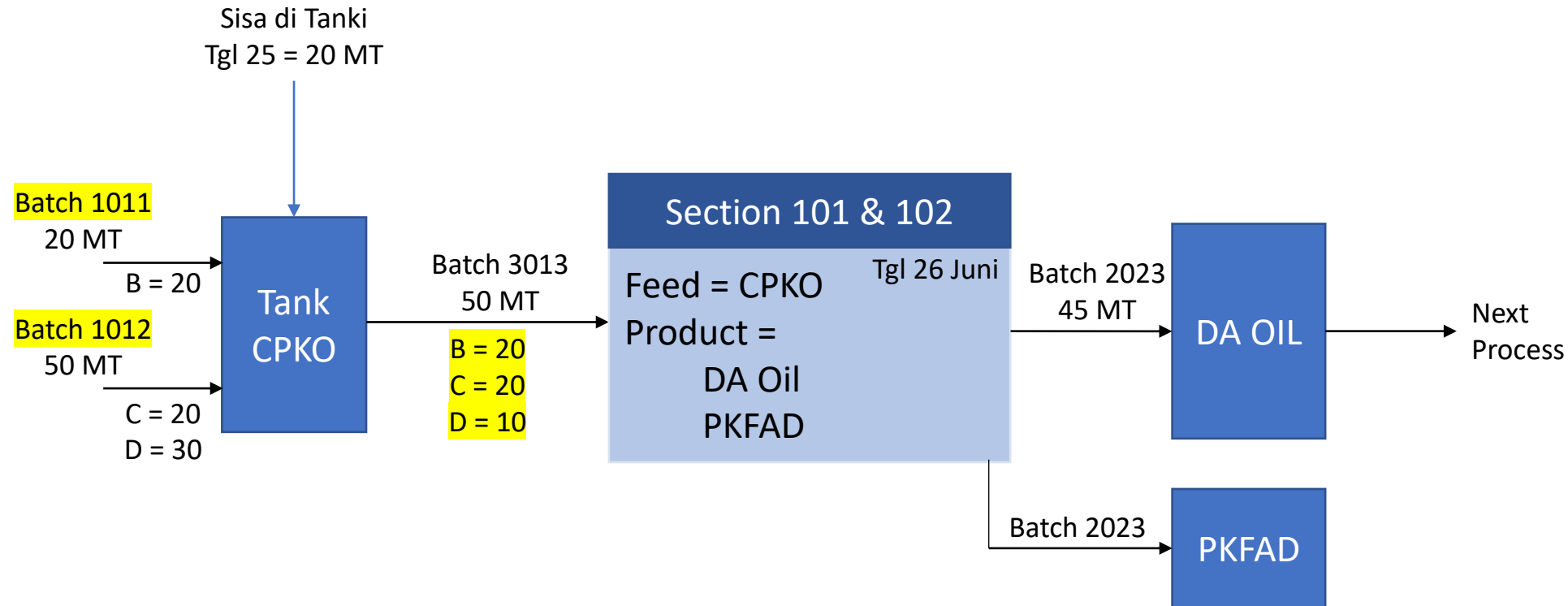
Delete Sequential in the Process



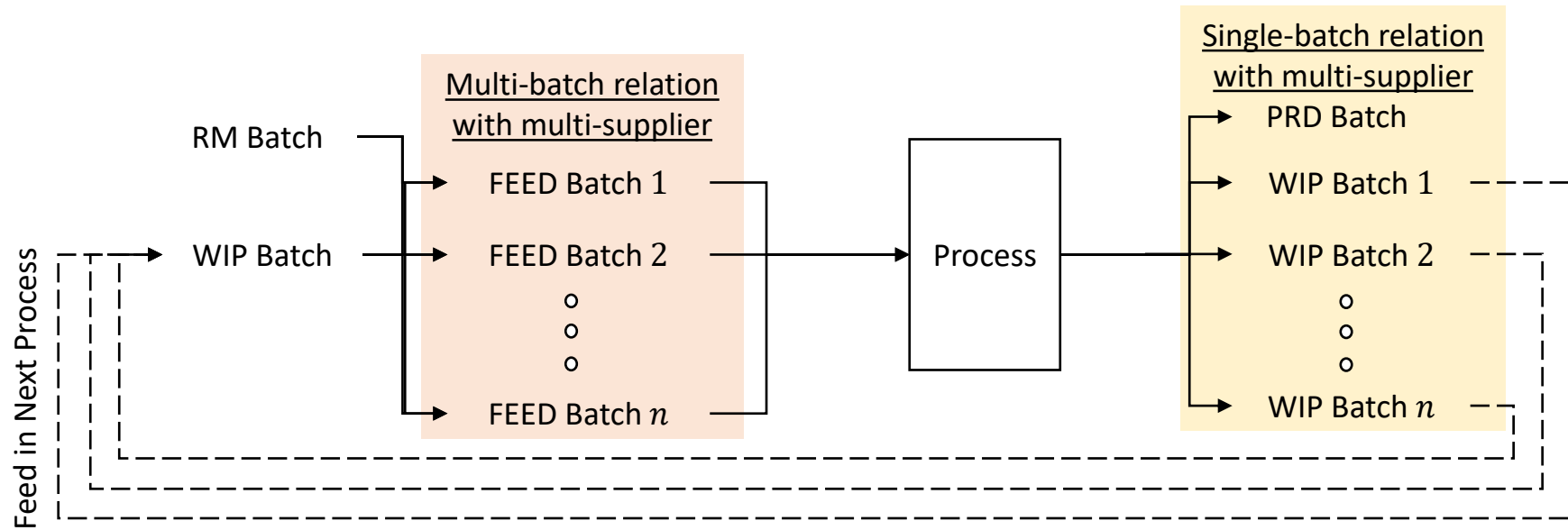
Example of Multi-Supplier



Example of Multi-Batch + Multi-Supplier



General Batch Relation



Eq. 1 →
$$\text{RM or WIP Batch (MT)} = \sum_{k=1}^n \text{FEED Batch } k \text{ (MT)}$$

Eq. 2 → **Process yield in each section (101 – 302)**

$$\text{Yield } k = \frac{\text{WIP Batch } k \text{ (MT)}}{\text{FEED Batch } k \text{ (MT)}} \times 100\%$$

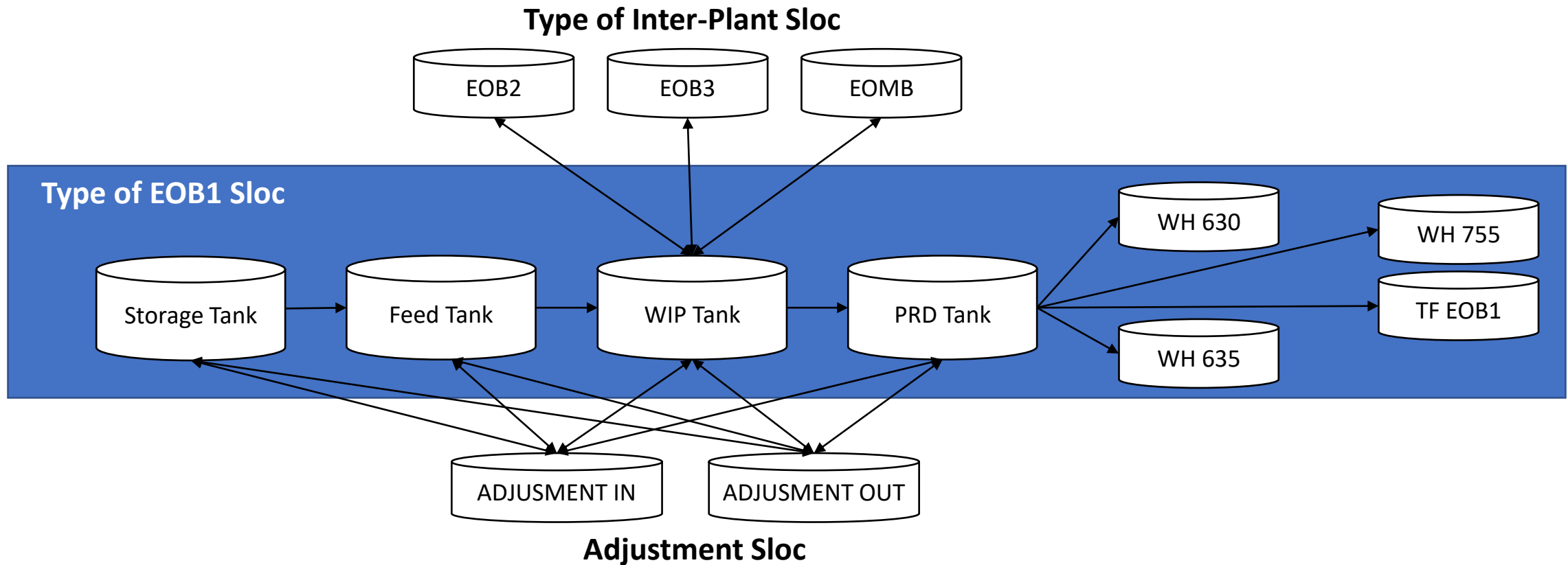
Eq. 2 → **Process yield in Transfer, Blending, Packaging**

$$\text{Yield } k = 100\%$$

Eq. 3 → Supplier in FEED Batch k → FIFO from RM or WIP Batch

Eq. 4 → Supplier in WIP Batch k → Process Yield k × Supplier in FEED Batch k

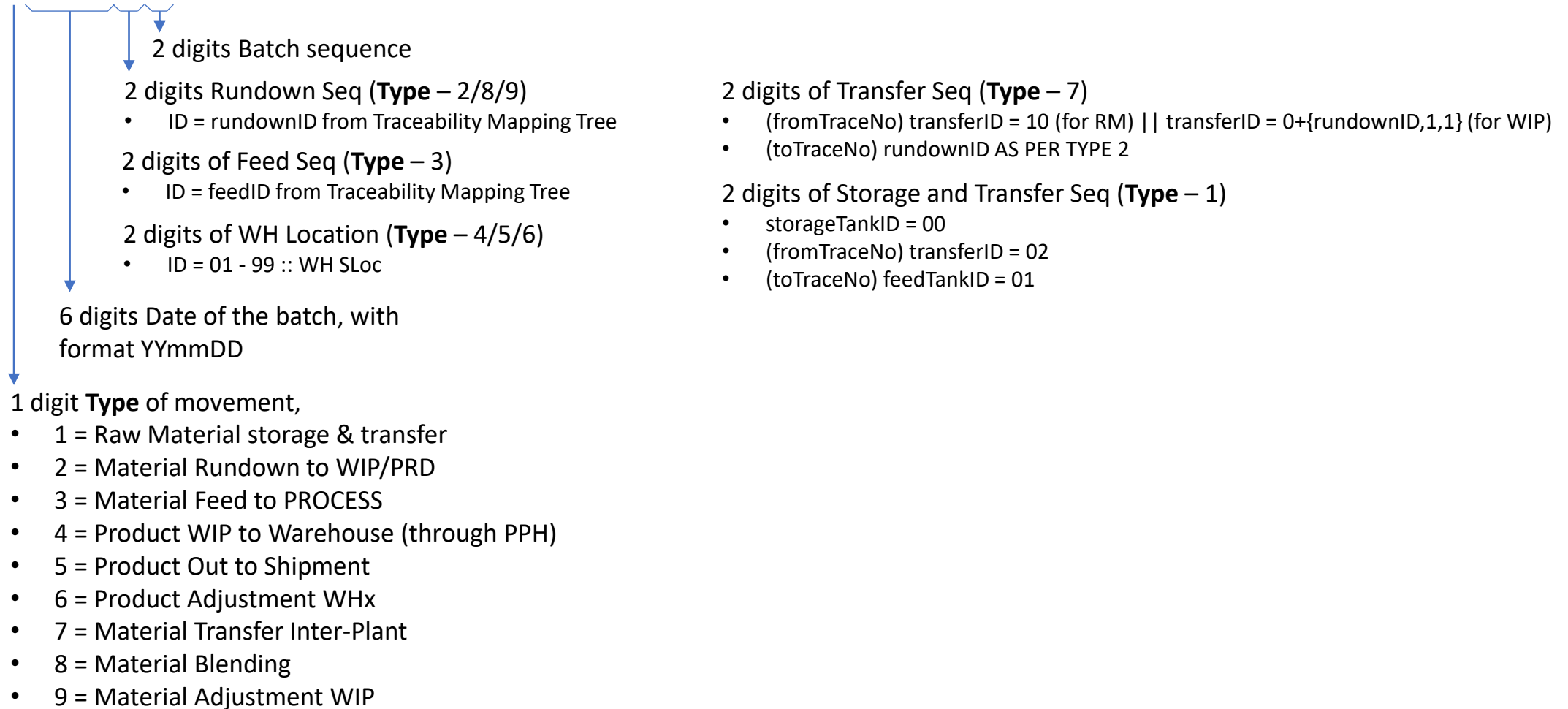
Storage Location (Sloc)



Batch Mapping System

- Trace number mapping system consists of 11 number formation

XXXXXXXXXX

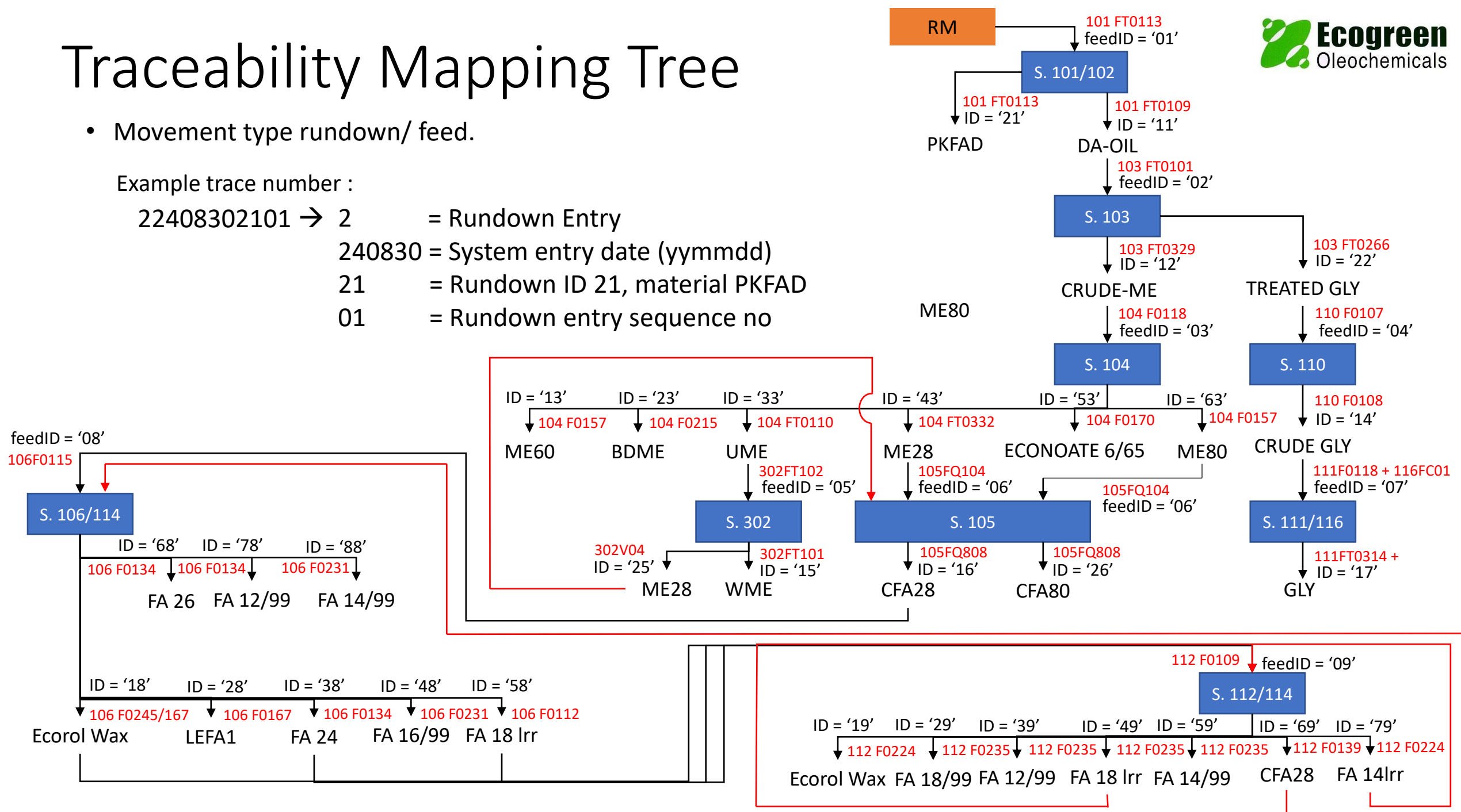


Traceability Mapping Tree

- Movement type rundown/ feed.

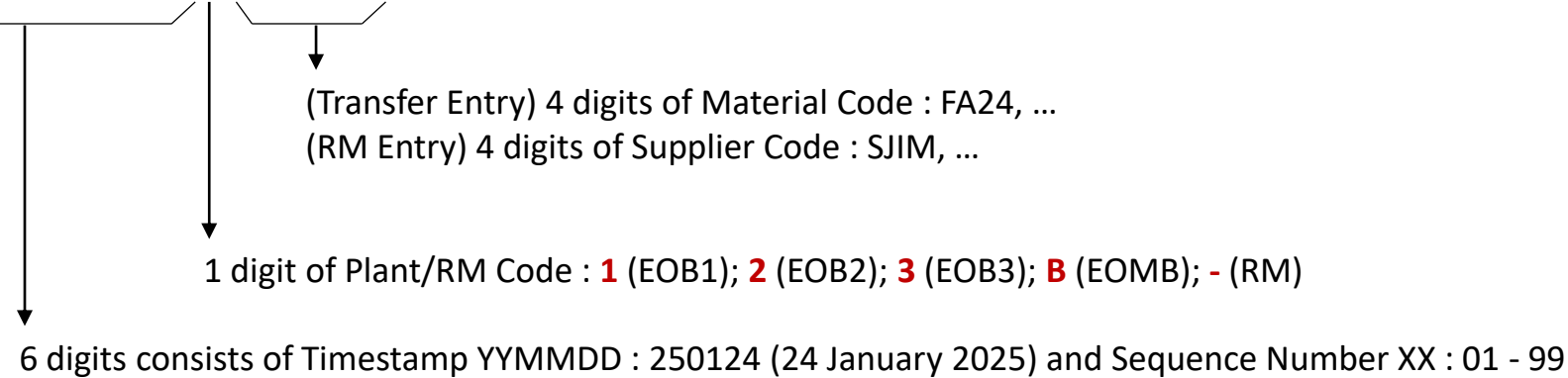
Example trace number :

22408302101 → 2 = Rundown Entry
 240830 = System entry date (yymmdd)
 21 = Rundown ID 21, material PKFAD
 01 = Rundown entry sequence no



Batch Traceability Standard Naming

YYMMDDXXPHHHH



Example (Transfer Entry):

- 250113011FA24 → timestamp 13 January 2025, sequence number 01, plant EOB1, material FA24
- 250113052FA24 → timestamp 13 January 2025, sequence number 05, plant EOB2, material FA24

Example (RM Entry):

- 25030601-SJIM → timestamp 6 March 2025, sequence number 01, raw material, supplier Sinar Jaya Inti Mulia

Raw Material Tracing System

2 digits of Storage and Transfer Seq (**Type** – 1)

- storageTankID = 00
- (fromTraceNo) transferID = 02
- (toTraceNo) feedTankID = 01

Example trace number :

IN to Storage Tank :: 12409120001

1 = Raw Material Entry
240912 = System entry date (yymmdd)
00 = IN to Storage Tank
01 = RM entry sequence number

OUT from Storage Tank :: 12409120201

1 = Raw Material Entry
240912 = System entry date (yymmdd)
02 = Transfer from Storage Tank to Feed Tank
01 = RM TRF entry sequence number

IN to Feed Tank :: 12409120101

1 = Raw Material Entry
240912 = System entry date (yymmdd)
01 = IN to Feed Tank
01 = RM TRF entry sequence number

Feed & Rundown Tracing System

2 digits Output Prd & Process Seq (**Type** – 2/8/9)

- ID = rundownID from Traceability Mapping Tree

2 digits of Transfer Seq (**Type** – 3)

- ID = feedID from Traceability Mapping Tree

Example trace number :

OUT from Feed Tank :: 32409130101

3 = Feed Entry

240913 = System entry date (yymmdd)

01 = Feed ID Sec 101/102 → RAW material CPKO/ Olein/ CPO

01 = Feed entry sequence number

IN to WIP Tank :: 22409131101

2 = Rundown Entry

240913 = System entry date (yymmdd)

11 = Rundown ID Sec 101/102 → WIP material DA-OIL

01 = Rundown entry sequence number

Packaging & Shipment Tracing System

id_warehouse	🔑 id_batch	code	description
1	P1	630	Warehouse 630
2	P2	630	Warehouse 630
3	13	630	Warehouse 630
4	14	630	Warehouse 630
5	1R	630	Warehouse 630
6	23	630	Warehouse 630
7	24	630	Warehouse 630
8	32	635	Warehouse 635
9	41	635	Warehouse 635
10	46	635	Warehouse 635
11	58	635	Warehouse 635
12	62	635	Warehouse 635
13	71	635	Warehouse 635
14	88	635	Warehouse 635
15	96	635	Warehouse 635
16	99	635	Warehouse 635
17	P3	755	Warehouse 755
18	P4	755	Warehouse 755
19	P5	755	Warehouse 755
20	P6	755	Warehouse 755
21	FB	630	Warehouse 630
22	IS	630	Warehouse 630
23	VS	122	Tank Farm EOB1

Example trace number :

OUT from WIP :: 42409120002

4 = Product WIP to Warehouse by Packaging Entry

240912 = System entry date (yymmdd)

00 = Transfer from WIP to Warehouse

02 = Packaging entry sequence number

IN to Warehouse :: 42409122302

4 = Product WIP to Warehouse by Packaging Entry

240912 = System entry date (yymmdd)

23 = Warehouse ID :: Batch VS, 122 – Tank Farm EOB1

02 = Packaging entry sequence number

OUT from Warehouse :: 52409182301

5 = Product Warehouse to Shipment by Shipment Entry

240918 = System entry date (yymmdd)

23 = Warehouse ID :: Batch VS, 122 – Tank Farm EOB1

01 = Shipment entry sequence number

Transfer Tracing System

2 digits of Transfer Seq (**Type** – 7)

- (fromTraceNo) transferID = 10 (for RM) || transferID = 0+{rundownID,1,1} (for WIP)
- (toTraceNo) rundownID AS PER TYPE 2

Example trace number :

OUT from Source Sloc :: 72409170301

7 = Transfer Entry
240917 = System entry date (yymmdd)
03 = Trf_ID
01 = Trf entry sequence number

IN to Destination Sloc :: 72409171301

7 = Transfer Entry
240917 = System entry date (yymmdd)
13 = Trf to RundownID 13, Sec 104, material ME60
01 = Trf entry sequence number

Blending Tracing System

2 digits Output Prd & Process Seq (**Type** – 2/8/9)

- ID = rundownID from Traceability Mapping Tree

Example trace number :

OUT from Source Sloc :: 82409040301

8 = Blending Entry
240904 = System entry date (yymmdd)
03 = Blending_ID
01 = Blending entry sequence number

IN to Destination Sloc :: 82409044301

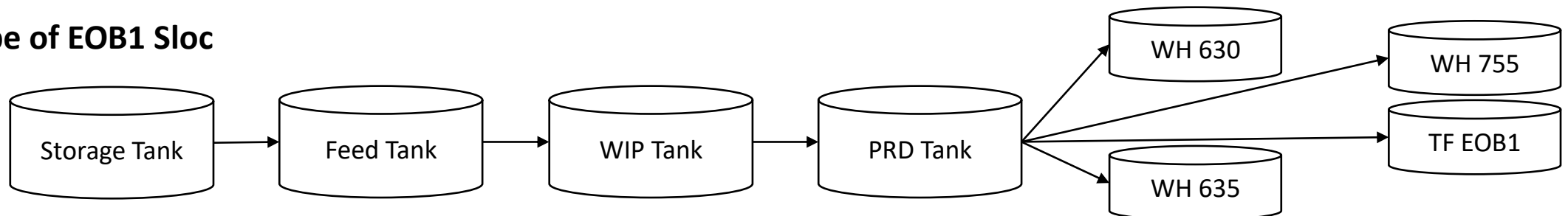
8 = Blending Entry
240904 = System entry date (yymmdd)
43 = Trf to RundownID 43, Sec 104, material ME28
01 = Trf entry sequence number

Trace Number Mapping System (1)

Transaction within EOB1 Sloc

	From Trace No	To Trace No	In Qty	Out Qty	Sloc
<u>Multi-batch relation with multi-supplier</u>	92408132919 Supplier A Supplier B	82409060001		55,00 35,00 20,00	WIP EOB1
	92409061004 Supplier A	82409060001		18,00 18,00	WIP EOB1
	92408291001 Supplier C	82409060001		80,00 80,00	WIP EOB1
<u>Single-batch relation with multi-supplier</u>	82409060001	82409061001 Supplier A Supplier B Supplier C	153,00 53,00 20,00 80,00		WIP EOB1

Type of EOB1 Sloc

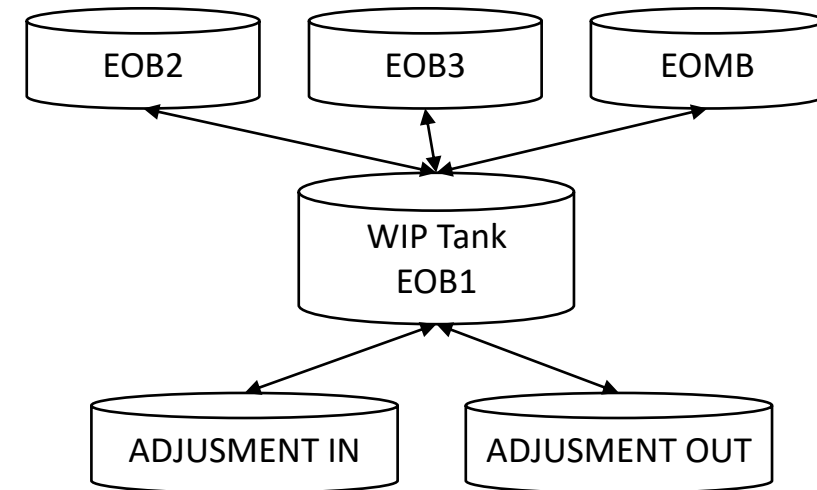


Trace Number Mapping System (2)

Transaction within several Sloc

From Trace No	To Trace No	In Qty	Out Qty	Sloc
22408281607 Supplier A Supplier B	72408300601		55,00 35,00 20,00	WIP EOB1
224082819606 Supplier A	72408300601		18,00 18,00	WIP EOB1
22408281606 Supplier C	72408300601		80,00 80,00	WIP EOB1
72408300601	72408301601	153,00		EOB2
	Supplier A	53,00		
	Supplier B	20,00		
	Supplier C	80,00		

Type of non-EOB1 Sloc



Multi-batch relation
with multi-supplier

Single-batch relation
with multi-supplier

Backward Tracing

SUMMARY

Trace No	SO No	Batch No	Sloc	Product Desc	Qty (MT)	Supplier / Batch SAP / Qty (MT)	Source Trace No / PO
52409170101	15423523	32-249-BT	635	ECOCEROL-PD250-DF	60.000	STOCK END JULY 2024 / 0731-DAOIL / Qty: 18.711 MT STOCK END JULY 2024 / 0731-GLY80 / Qty: 0.230 MT STOCK END JULY 2024 / 0731-TGW / Qty: 58.884 MT	0731-DAOIL :: 92408131101 / 4200006793
Shipment trace number	Sales Order	Batch Production	Warehouse	Product Name	Qty Shipment	Source / Batch SAP / Quantity in One Shipment	Batch SAP :: source Trace No / PO supplier

Backward Tracing

DETAIL TRACING

Path	↑↓	Curr Batch	↑↓	Prev Batch	↑↓	Batch Date	↑↓	Material	↑↓	RM/WIP/WHx In_Qty (MT)	↑↓	SLoc	↑↓	RM/WIP/WHx Out_Qty (MT)	↑↓	Supplier / Batch SAP / Init Qty (MT) / Remark	↑↓
1		52409170101		42409110811		2024-09-17		ECOCEROL-PD250-DF		0		WAREHOUSE 635		60		STOCK END JULY 2024 / 0731-DAOIL / 10.814 MT STOCK END JULY 2024 / 0731-TGW / 0.581 MT STOCK END JULY 2024 / 0731-GLY80 / 0.115 MT STOCK END JULY 2024 / 0731-TGW / 17.710 MT STOCK END JULY 2024 / 0731-DAOIL / 7.897 MT STOCK END JULY 2024 / 0731-TGW / 22.883 MT	
1.03		42409110811		42409110011		2024-09-06		ECOCEROL-PD250-DF		60		WAREHOUSE 635		0		STOCK END JULY 2024 / 0731-DAOIL / 10.814 MT STOCK END JULY 2024 / 0731-TGW / 0.581 MT STOCK END JULY 2024 / 0731-GLY80 / 0.115 MT STOCK END JULY 2024 / 0731-TGW / 17.710 MT STOCK END JULY 2024 / 0731-DAOIL / 7.897 MT STOCK END JULY 2024 / 0731-TGW / 22.883 MT	
1.03.01		42409110011		22408271706		2024-09-06		ECOCEROL-DF		0		EOB1 PRODUCT TANK		17.825		STOCK END JULY 2024 / 0731-GLY80 / 0.115 MT STOCK END JULY 2024 / 0731-TGW / 17.710 MT	
1.03.01.01		22408271706		32408270706		2024-08-07		ECOCEROL-DF		28.41				0		STOCK END JULY 2024 / 0731-GLY80 / 0.115 MT STOCK END JULY 2024 / 0731-TGW / 28.295 MT	

Monthly Adj. (PSPA vs On-Hand)

- Go to [Adjustment] >> [WIP Adjustment] >> then click [Period Adjustment] button

PTEO EUDR-TS

DASHBOARD

» Forward Trace

« Backward Trace

TS TRANSACTION

Raw Material

WIP

Blending

Transfer

Packaging

Shipment

TS INQUIRY

TS Report

Stock On-Hand

TS SETUP

Adjustment **1**

Adjustment - WIP

2

WIP Adjustment WAREHOUSE Adjustment

3

Adjustment Last Record Stock Initialization Period Adjustment

Show 10 entries

No	Action	Adjustment No	Material No	Entry Date	Material
1		92409300001	Add Doc No	2024-09-30	50000100106010 :: CPKO-DF
2		92409300002	Add Doc No	2024-09-30	50000100106010 :: CPKO-DF
3		92409300003	Add Doc No	2024-09-30	50000100106010 :: CPKO-DF
4		92409300004	Add Doc No	2024-09-30	50000900006010 :: CRUDE PALM KERNEL OLEIN
5		92409301006	Add Doc No	2024-	5102000106010

- For new adjustment period >> click [Create New Period] button

Stock Period Adjustment Entry

4

Create New Period

Show 10 entries

Search:

No	Action	Period	Batch	Uploaded File	Adjustment Status	Created At	Updated At
1		2024-09-30	STOCK END SEPT 2024	✓	✗	2024-09-26 11:20:33	2024-09-26 12:54:25
2		2024-08-31	STOCK END AUGUST 2024	✗	✗	2024-09-26 10:22:07	2024-09-26 11:20:08

Showing 1 to 2 of 2 entries

Previous **1** Next

Monthly Adj. (PSPA vs On-Hand)

- Fill in end of period date, batch SAP then save new Period Adjustment.

New Adjustment Period

Entry Mode

ADD

Period

5

31/10/2024

Batch SAP

6

STOCK END OCT 2024

7

Save

- Upload excel file by click green button. After upload, cannot delete Period data.

Stock Period Adjustment Entry

Create New Period

Show 10 entries

Search:

No	Action	Period	Batch	Uploaded File	Adjustment Status	Created At	Updated At
1	8	2024-10-31	STOCK END OCT 2024	×	×	2024-09-30 11:16:49	

- Get file in the designated folder, then save.

Upload Excel

Get file

Choose File

No file chosen

Save

9

[NOTE] → excel file format must follow template

Monthly Adj. (PSPA vs On-Hand)

- Excel file format. Must consist of 10 columns start from Column A until Column J. Uploaded data must follow according to the designated column header name.
- Column sequential data must follow according to the below example.

	A	B	C	D	E	F	G	H	I	J
1	PlantSour	Tank	Material	Material Description 1	Capac	Sound	Temperatu	Volur	Dens	End Qty (N
2	1002	122T08	5303000006010	LIGHT END FATTY ALCOHOL 1	150,000	299,000	33,000	80,144	0,814	65,261
3	1002	122T09	5105000006010	METHYL ESTER CRUDE	300,000	329,100	75,000	119,342	0,828	98,815
4	1002	122T10	5105000006010	METHYL ESTER CRUDE	150,000	250,200	28,000	72,080	0,866	62,392
5	1002	122T11	5105280006010	METHYL ESTER 28	1.200,000	311,500	50,000	484,761	0,846	410,205
6	1002	122T12	5105280006010	METHYL ESTER 28	1.200,000	161,800	39,000	268,885	0,855	229,897
7	1002	122T13	5205180106010	ECONOATE 18 U	600,000	77,600	30,000	67,172	0,870	58,426
8	1002	122T13A	5205180106010	ECONOATE 18 U	600,000	212,000	50,000	167,150	0,856	143,047
9	1002	122T15	5105600006010	METHYL ESTER 60	2.400,000	546,100	38,000	1.054,489	0,860	906,650
10	1002	122T16	5103280006010	ALCOHOL CRUDE 28	1.200,000	401,000	58,000	597,887	0,809	483,451

Monthly Adj. (PSPA vs On-Hand)

- (10) View uploaded file and do adjustment by clicking blue button [View Detail].

Stock Period Adjustment Entry

[Create New Period](#)

Show entries

No	Action	Period	Batch	Uploaded File
1	 	2024-10-31	STOCK END OCT 2024	

10

- (11) Re-upload if there is revised of data in the Excel file, before Adjustment.
- (12) Then, calculate Qty On-Hand by end of Period >> in this example until 31 October 2024 (Note :: finish all entry by 31/10/2024, before calculate).

View Uploaded Data & Adjustment

Period: Batch SAP:

[0. Re-upload Excel](#) [1. Calc Qty \(On-Hand\)](#) [2. Do Adjustment](#)

Show entries **11** **12**

Search:

No	Plant	Tank	Sloc	Material	Qty (PSPA)	Qty (On-Hand)	Total	Adj.Type	Adj.No	Adj.Status	Created At	Updated At
1	1002	504T02	FEED TANK	CPKO (5000010006010)	2,140.462	0.000	-2,140.462			?	2024-09-30 11:30:09	
2	1002	504T09	STORAGE TANK	CPKO (5000010006010)	5,793.994	0.000	-5,793.994			?	2024-09-30 11:30:09	

Monthly Adj. (PSPA vs On-Hand)

- After On-Hand calc, Adj.Type has been defined, check again overall data.
- Continue with Adjustment by click button [Do Adjustment]

View Uploaded Data & Adjustment

Period

31/10/2024

Batch SAP

STOCK END OCT 2024

0. Re-upload Excel

1. Calc Qty (On-Hand)

2. Do Adjustment

Show

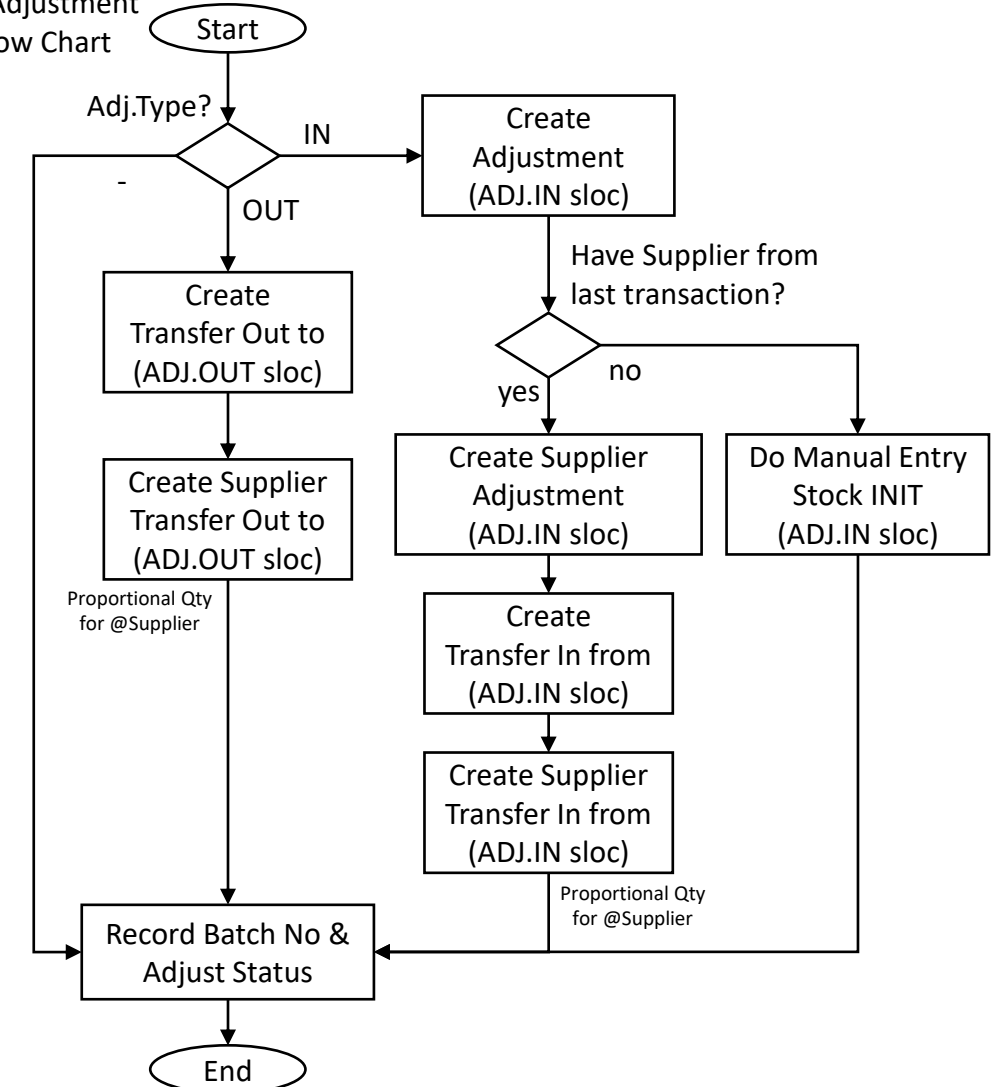
10

entries

13

No	Plant	Tank	Sloc	Material	Qty (PSPA)	Qty (On-Hand)	Total	Adj.Type
1	1002	504T02	FEED TANK	CPKO (5000010006010)	2,140.462	948.323	-1,192.139	IN
2	1002	504T09	STORAGE TANK	CPKO (5000010006010)	5,793.994	0.000	-5,793.994	IN

Do Adjustment Flow Chart



Monthly Adj. (PSPA vs On-Hand)

- Repeat [Do Adjustment] until all adjustment status is CHECKED (green status).

View Uploaded Data & Adjustment

Period

Batch SAP

31/10/2024

STOCK END OCT 2024

0. Re-upload Excel

1. Calc Qty (On-Hand)

2. Do Adjustment

Show

10

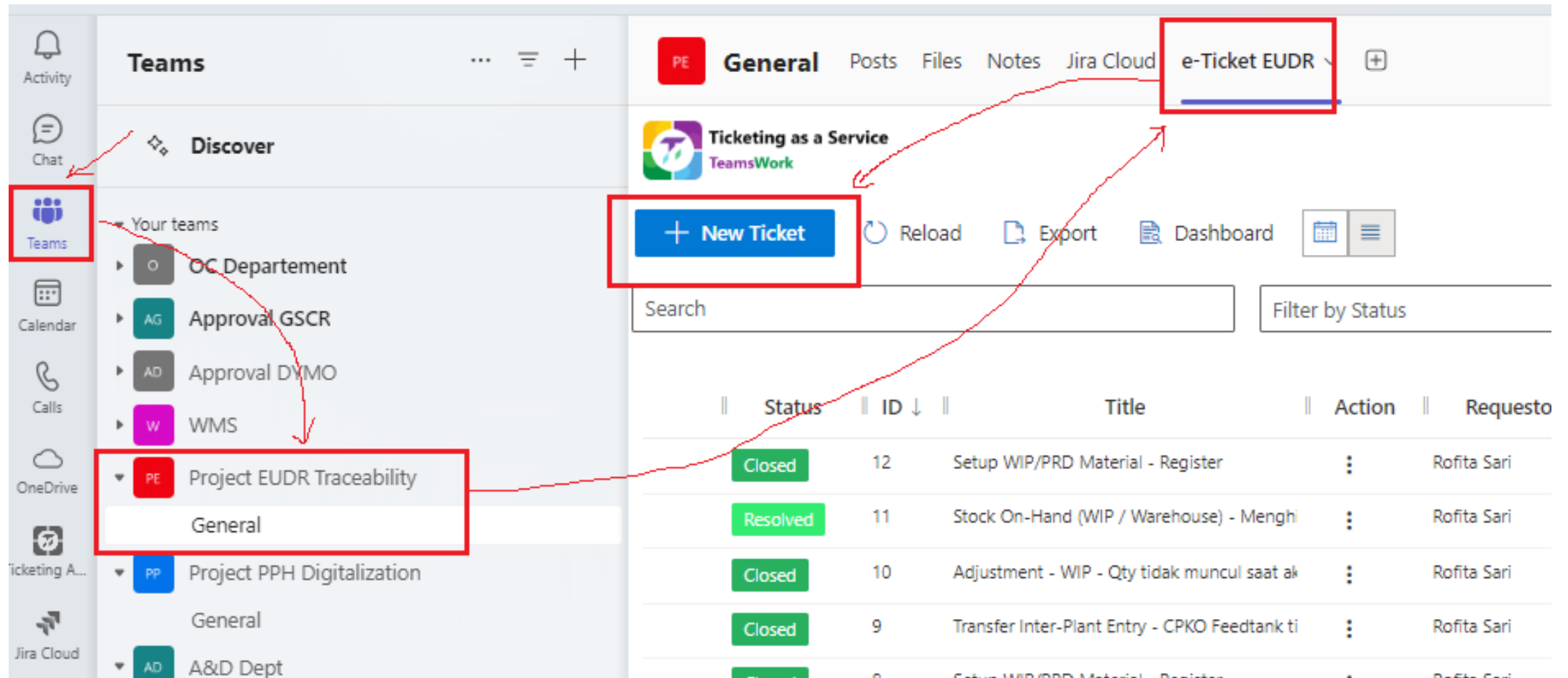
entries

Search:

No	Plant	Tank	Sloc	Material	Qty (PSPA)	Qty (On-Hand)	Total	Adj.Type	Adj.No	Adj.Status
1	1002	504T02	FEED TANK	CPKO (5000010006010)	2,140.462	948.323	-1,192.139	IN	92409301024 >>> 72410310001	✓
2	1002	504T09	STORAGE TANK	CPKO (5000010006010)	5,793.994	0.000	-5,793.994	IN	92409300025 >>> 72410310002	✓
3	1002	504T15	STORAGE TANK	CPO (5000050006010)	300.291	0.000	-300.291	IN	Manual Stock Init	✗

E-Ticket for Request Traceability System

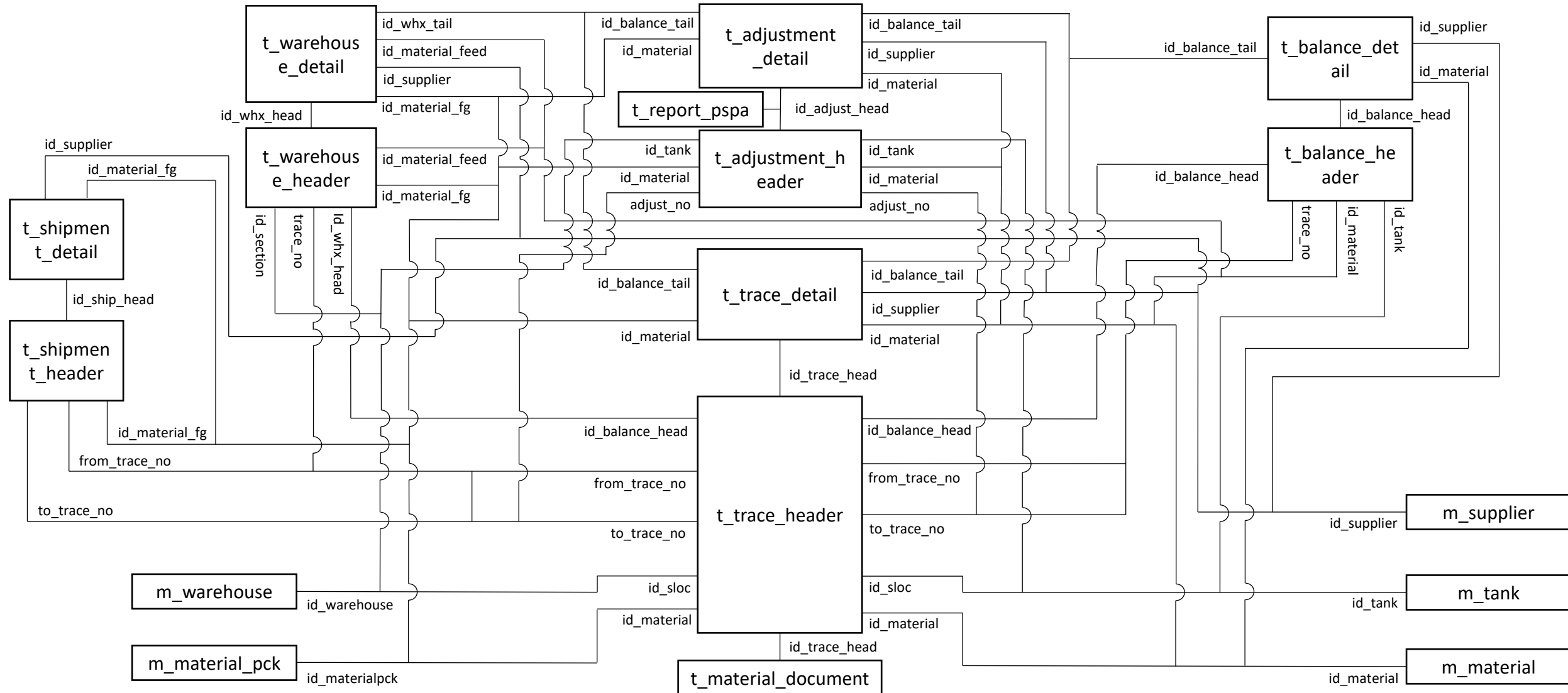
Please send request regarding the application through e-ticketing system in Teams.



The screenshot displays the Microsoft Teams interface with the 'e-Ticket EUDR' system. The left sidebar shows the 'Teams' icon highlighted. The main area shows the 'Project EUDR Traceability' team, with the 'General' channel selected. A 'New Ticket' button is visible in the channel header. The 'e-Ticket EUDR' tab is active, showing a list of tickets.

Status	ID	Title	Action	Requestor
Closed	12	Setup WIP/PRD Material - Register	⋮	Rofita Sari
Resolved	11	Stock On-Hand (WIP / Warehouse) - Mengh	⋮	Rofita Sari
Closed	10	Adjustment - WIP - Qty tidak muncul saat ak	⋮	Rofita Sari
Closed	9	Transfer Inter-Plant Entry - CPKO Feedtank ti	⋮	Rofita Sari
Closed	8	Setup WIP/PRD Material - Register	⋮	Rofita Sari

Design Overview



Trace Number Generation

- Raw Material – Storage Tank Module

```
static function get_rmNewEntryNumber()  
{  
    $db = DB::select('SELECT a.rm_number  
                    FROM (SELECT a.trace_no+1 AS rm_number  
                        FROM t_balance_header a  
                        WHERE SUBSTRING(a.trace_no,1,7) = CONCAT("1", DATE_FORMAT(CURDATE(), "%y%m%d"))  
                        AND SUBSTRING(a.trace_no,8,2) = ?  
                        AND a.status = 1  
                        ORDER BY a.id_balance_head DESC  
                        LIMIT 1 ) a  
                    UNION ALL  
                    SELECT CONCAT("1", DATE_FORMAT(CURDATE(), "%y%m%d"), ?, "01") AS rm_number  
                    LIMIT 1', [self::$movSeq, self::$movSeq]);  
    return $db;  
}
```

You, 3 months ago • ver20240615_12:28_sw

```
protected static $idTankSrc = "T00"; // STORAGE TANK  
protected static $idTankTrf = "T02"; // TRF TANK  
protected static $idTankFeed = "T01"; // FEED TANK  
protected static $movSeq = "00";  
protected static $typeMaterial = "RM";  
protected static $movType1 = "1";  
protected static $movType2 = "9";  
protected static $idPlantEob1 = "1002";
```


Trace Number Generation

- Raw Material – Feed Tank Module

```
static function get_rmNewEntryNumberTrf(){
    $db = DB::select('SELECT CONCAT(SUBSTRING(a.rm_number,1,7), ?, SUBSTRING(a.rm_number,10,2)) + 1 AS rm_number
    FROM (SELECT a.trace_no AS rm_number
    FROM t_balance_header a
    WHERE SUBSTRING(a.trace_no,1,7) = CONCAT("1", DATE_FORMAT(CURDATE(), "%y%m%d"))
    AND SUBSTRING(a.trace_no,2,8) = CONCAT(DATE_FORMAT(CURDATE(), "%y%m%d"), ?)
    AND a.status = 1
    ORDER BY a.id_balance_head DESC
    LIMIT 1 ) a
    UNION ALL
    SELECT CONCAT("1", DATE_FORMAT(CURDATE(), "%y%m%d"), ?, "01") AS rm_number
    LIMIT 1', [substr(self::$idTankSrc,1,2), substr(self::$idTankFeed,1,2), substr(self::$idTankSrc,1,2)]);
    return $db;
}
```

```
protected static $idTankSrc = "T00"; // STORAGE TANK
protected static $idTankTrf = "T02"; // TRF TANK
protected static $idTankFeed = "T01"; // FEED TANK
protected static $movSeq = "00";
protected static $typeMaterial = "RM";
protected static $movType1 = "1";
protected static $movType2 = "9";
protected static $idPlantEob1 = "1002";
```

Trace Number Generation

- WIP – Feed Module

```
static function get_feedNewBatchNumber($request){
    $feedID = $request->input('feedID');

    $feedID = substr($feedID, 0, 2);
    $db = DB::select('SELECT a.feed_number
                    FROM (SELECT a.to_trace_no+1 AS feed_number
                        FROM t_trace_header a
                        WHERE SUBSTRING(a.to_trace_no,1,9) = CONCAT(3, DATE_FORMAT(CURDATE(), "%y%m%d"), ?)
                        AND a.status = 1
                        ORDER BY a.id_trace_head DESC
                        LIMIT 1 ) a
                    UNION ALL
                    SELECT CONCAT(3, DATE_FORMAT(CURDATE(), "%y%m%d"), ? , "01") AS feed_number
                    LIMIT 1', [$feedID, $feedID]);

    return $db;
}
```

Trace Number Generation

- WIP – Rundown Module (1)

```
static function get_rundownNewBatchNumber($request){
    $rundownID = $request->input('rundownID');
    $section = substr($rundownID, 1, 1);

    if ($section == 9){
        /* GET LATEST FEED TRACE NO */
        $db = DB::select('SELECT CONCAT(2, SUBSTRING(a.to_trace_no,2,6), ?, SUBSTRING(a.to_trace_no,10,2)) AS rundown_number
                        FROM (SELECT to_trace_no
                              FROM t_trace_header a
                              WHERE a.status = 1
                                   AND SUBSTRING(a.to_trace_no,1,9) = CONCAT(3, DATE_FORMAT(CURDATE(), "%y%m%d"), "09")
                              ORDER BY to_trace_no DESC
                              LIMIT 1) a
                        UNION ALL
                        SELECT CONCAT(2, DATE_FORMAT(CURDATE(), "%y%m%d"), ? , "01") AS rundown_number
                        LIMIT 1', [$rundownID, $rundownID]);
    } elseif ($section == 8){
```

Trace Number Generation

- WIP – Rundown Module (2)

```
} elseif ($section == 8){  
    $db = DB::select('SELECT CONCAT(2, SUBSTRING(a.to_trace_no,2,6), ?, SUBSTRING(a.to_trace_no,10,2)) AS rundown_number  
                    FROM (SELECT to_trace_no  
                        FROM t_trace_header a  
                        WHERE a.status = 1  
                        AND SUBSTRING(a.to_trace_no,1,9) = CONCAT(3, DATE_FORMAT(CURDATE(), "%y%m%d"), "08")  
                        ORDER BY to_trace_no DESC  
                        LIMIT 1) a  
                    UNION ALL  
                    SELECT CONCAT(2, DATE_FORMAT(CURDATE(), "%y%m%d"), ? , "01") AS rundown_number  
                    LIMIT 1', [$rundownID, $rundownID]);  
}  
else {
```

Trace Number Generation

- WIP – Rundown Module (3)

```
} else {  
    $db = DB::select('SELECT a.rundown_number  
                      FROM (SELECT a.to_trace_no+1 AS rundown_number  
                            FROM t_trace_header a  
                            WHERE SUBSTRING(a.to_trace_no,1,9) = CONCAT(2, DATE_FORMAT(CURDATE(), "%y%m%d"), ?)  
                            AND a.status = 1  
                            ORDER BY a.id_trace_head DESC  
                            LIMIT 1 ) a  
                      UNION ALL  
                      SELECT CONCAT(2, DATE_FORMAT(CURDATE(), "%y%m%d"), ? , "01") AS rundown_number  
                      LIMIT 1', [$rundownID, $rundownID]);  
}  
return $db;  
}
```

Trace Number Generation

- WIP – Blending Module

```
$db = DB::select('SELECT b.trace_no + 1 AS entryNo
FROM m_material a
LEFT JOIN t_balance_header b
ON a.id_rundown = SUBSTRING(b.trace_no, 8,2) AND b.status = 1
WHERE a.id_material = ?
AND SUBSTRING(b.trace_no, 1, 7) = CONCAT(8, DATE_FORMAT(CURDATE(), "%y%m%d"))
AND a.status = 1
UNION ALL
SELECT CONCAT("8", DATE_FORMAT(CURDATE(), "%y%m%d"), IF(a.id_rundown <> "-", a.id_rundown, "00"), "01") AS entryNo
FROM m_material a
WHERE a.status = 1
AND a.id_material = ?
LIMIT 1
', [$idMaterial, $idMaterial]);
```

Trace Number Generation

- WIP – Transfer Module

```
static function get_newTransferEntryNo($request){
    $idMaterial = $request->input('id_material');

    $db = DB::select('SELECT a.entryNo
        FROM (SELECT CONCAT(7,DATE_FORMAT(CURDATE(), "%y%m%d"),a.id_rundown, LPAD(SUBSTRING(b.trace_no,10,2) + 1,2,0)) AS entryNo
        FROM m_material a
        LEFT JOIN t_balance_header b
        ON SUBSTRING(a.id_rundown,2,1) = SUBSTRING(b.trace_no, 9,1) AND b.status = 1
        WHERE a.id_material = ?
        AND SUBSTRING(b.trace_no, 1, 1) = 7
        AND SUBSTRING(b.trace_no, 2, 6) = DATE_FORMAT(CURDATE(), "%y%m%d")
        AND a.status = 1
        ORDER BY SUBSTRING(b.trace_no,10,2) DESC
        LIMIT 1) a
        UNION ALL
        SELECT CONCAT("7", DATE_FORMAT(CURDATE(), "%y%m%d"), IF(a.id_rundown <> "-", a.id_rundown, "00"), "01") AS entryNo
        FROM m_material a
        WHERE a.status = 1
        AND a.id_material = ?
        LIMIT 1
        ', [$idMaterial, $idMaterial]);

    return $db;
}
```

Trace Number Generation

- Warehouse – Packaging Module

```
/* CREATE BATCH NUMBER */
$datPckBatch = DB::select('SELECT a.pck_batch
FROM (SELECT CONCAT(?, DATE_FORMAT(CURDATE(), "%y%m%d"), ?, LPAD(SUBSTRING(a.to_trace_no,10,2) + 1,2,0)) AS pck_batch
FROM t_trace_header a
WHERE SUBSTRING(a.to_trace_no,1,7) = CONCAT(?, DATE_FORMAT(CURDATE(), "%y%m%d"))
AND a.status = 1
ORDER BY a.id_trace_head DESC
LIMIT 1 ) a
UNION ALL
SELECT CONCAT(?, DATE_FORMAT(CURDATE(), "%y%m%d"), ? , "01") AS pck_batch
LIMIT 1', [self::$movType1, $whID, self::$movType1, self::$movType1, $whID]);
$traceNoWhx = $datPckBatch[0]->pck_batch;
// Problem with Trace Loop
$traceNoTrf = substr_replace($traceNoWhx, '00', 7, 2); /* REPLACE WHX_ID WITH TRANSFER_ID */
```


Trace Number Generation

- Warehouse – Shipment Module

```
/* CREATE SHIPMENT BATCH NUMBER */
$datPckBatch = DB::select('SELECT a.pck_batch
FROM (SELECT a.to_trace_no + 1 AS pck_batch
FROM t_trace_header a
WHERE SUBSTRING(a.to_trace_no,1,9) = CONCAT(5, DATE_FORMAT(CURDATE(), "%y%m%d"), ?)
AND a.status = 1
ORDER BY a.id_trace_head DESC
LIMIT 1 ) a
UNION ALL
SELECT CONCAT(5, DATE_FORMAT(CURDATE(), "%y%m%d"), ? , "01") AS pck_batch
LIMIT 1', [$shID, $shID]);
$traceNo = $datPckBatch[0]->pck_batch;
```

Trace Number Generation

- WIP – Adjustment Module

```
static function get_adjNewEntryNumber(){
    $db = DB::select('SELECT a.adj_number
                    FROM (SELECT a.trace_no+1 AS adj_number
                        FROM t_balance_header a
                        WHERE SUBSTRING(a.trace_no,1,7) = CONCAT("9", DATE_FORMAT(CURDATE(), "%y%m%d"))
                        AND a.status = 1
                        ORDER BY a.id_balance_head DESC
                        LIMIT 1 ) a
                    UNION ALL
                    SELECT CONCAT("9", DATE_FORMAT(CURDATE(), "%y%m%d"), "0001") AS adj_number
                    LIMIT 1');
    return $db;
}
```

Trace Number Generation

- Warehouse – Adjustment Module

```
static function get_adjNewEntryNumberWhx(){
    $db = DB::select('SELECT a.adj_number
                    FROM (SELECT a.trace_no+1 AS adj_number
                        FROM t_warehouse_header a
                        WHERE SUBSTRING(a.trace_no,1,7) = CONCAT("6", DATE_FORMAT(CURDATE(), "%y%m%d"))
                        AND a.status = 1
                        ORDER BY a.id_whx_head DESC
                        LIMIT 1 ) a
                    UNION ALL
                    SELECT CONCAT("6", DATE_FORMAT(CURDATE(), "%y%m%d"), "0001") AS adj_number
                    LIMIT 1');
    return $db;
}
```